



Label adversarial domain adaptation network for predicting remaining useful life based on cross-domain condition[☆]

Shanshan Lv^a, Chengcheng Xia^a, Cong Cheng^a, Jianhai Yan^b, Xiaodan Wu^{a,*}

^a School of Economics and Management, Hebei University of Technology, Tianjin, China

^b Business School, University of Shanghai for Science and Technology, Shanghai, China

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ABSTRACT

Most data-driven methods for predicting remaining useful life assume that the data under different operating conditions follow the same distribution. However, this assumption rarely holds in real-world situation. Additionally, traditional methods do not fully utilize the hidden label information from the target domain or account for the transfer quality of source domain data. To address these issues, Label Adversarial Domain Adaptation (LADA) network is introduced in this paper. Specifically, LADA aims to filter the source domain data and maximize the use of hidden label information from the target domain. Firstly, a similarity measurement indicator based on the pearson correlation coefficient (PCC) and dynamic time warping (DTW) is employed to filter source domain data similar to the target domain data distribution. Then, in order to fully utilize the hidden label information from the target domain, the cloud model and golden section are utilized to create pseudo class labels. Furthermore, a feature difference module is established that minimizes the disparity between domain features. This is realized by using the maximum mean difference (MMD) and Kolmogorov–Smirnov (K–S) statistical test. The experimental results indicate that LADA has advantages in cross-domain RUL prediction.

1. Introduction

Remaining useful life (RUL) prediction, being an essential part of Prognostic and Health Management (PHM), is core for ensuring efficient, safe and reliable mechanical equipment operations (Yang, Liu, & Zio, 2019). Generally, RUL prediction techniques can be roughly classified into two categories (Zhang et al., 2021). One is the model-based approach, the other is the data-driven method. The former estimates RUL by physical or mathematical modeling (Jouin et al., 2016). However, their accuracy is often limited to the complexity of operating mechanisms and necessity for prior equipment knowledge. Conversely, data-driven methods bypass the need for prior equipment knowledge and do well in extracting deep degradation information from data.

In practical situations, the working conditions of mechanical equipment always change. Owing to changes in the working conditions of the equipment, there are widespread differences in data distribution, namely domain shift. Therefore, data-driven methods based on the

assumption of the same data distribution are not suitable for cross-domain RUL prediction. Transfer learning (TL) can effectively solve the problem of domain shift by applying the knowledge learned from the source domain to the target domain (Jia et al., 2022). Domain adaptation (DA) is combined with long short-term memory network (LSTM) to predict the RUL of unlabeled target domain data (da Costa et al., 2020). CORAL (Sun, Feng, & Saenko, 2017) can minimize domain shift to the maximum extent without requiring any target labels. Shen et al. (2018) introduced the Wasserstein distance into DA networks and designed a new feature representation method. Learning domain invariant features which are semantically meaningful can help DA minimize the domain shift (Tzeng et al., 2014).

As a branch of TL, domain adversarial neural networks (DANN) have achieved satisfactory performance in aligning the distribution differences between two domains. For example, DANN approach has been used to adaptively estimate RUL of unlabeled target domain (Zhang

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* Correspondence to: School of Economics and Management, Hebei University of Technology, No. 5340 Xiping Rd., Tianjin 300401, PR China.

E-mail addresses: shanshanlv@tju.edu.cn (S. Lv), xiacc8057@163.com (C. Xia), ccheng@hebut.edu.cn (C. Cheng), jianhai@usst.edu.cn (J. Yan), xwu@hebut.edu.cn (X. Wu).

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et al., 2018). Ragab et al. (2020) proposed an approach with a contrastive loss based on adversarial domain adaptation network to consider target-specific information. Wu et al. (2022) proposed a weight adversarial loss to align samples with similar RULs from the source and target domains. Tzeng et al. (2017) used a loss based generative adversarial network (GAN) to determine the target characteristics matched with the source characteristics. To address the problem of different machine tools that had different service time, a weighted adaptive joint distribution adaptation TL method was proposed to predict the RUL related to position-speed (Li et al., 2022). Wang et al. (2023) proposed a capacity estimation approach based on DA using feature disentanglement and tendency retainment network. Zhuang, Jia, and Zhao (2022) developed an adversarial domain adaptation architecture using supervised positive contrast modules. Hu et al. (2022) used the deep feature disentanglement TL networks to collect domain invariant features. Ye and Yu (2022) proposed selective adversarial adaptive networks to realize knowledge selection and updates selective weights based on different data distributions. The above-mentioned methods have achieved excellent performance by decreasing the data distribution discrepancy between two domains. However, few studies have considered the transferability of source domain data.

Transferability means the generalization ability and universality of a model across different data domains or datasets. The transferability of source domain data also affects the effectiveness of transfer learning (Zhong et al., 2021). However, most current research on RUL prediction based on transfer learning rarely considers the transferability of source domain data. By integrating distribution differences such as KL divergence (Tian et al., 2023) or MMD into the network, the alignment between the source and target datasets can be achieved. However, most of the above metrics are only designed for marginal distribution adaptation (Liu et al., 2021). Wang and Tsung (2022) proposed a joint prior function for cross-product data to explicitly capture the similarity between degradation data of two different products. Cheng et al. (2023) used reverse validation techniques to select appropriate source samples to avoid the negative transfer caused by inappropriate source samples affecting model performance. The above methods have achieved satisfactory performance in measuring differences in data distribution. However, they overlooked the impact of the transferability of source domain data on transfer learning performance.

The target domain data is usually truncated in practical industrial but still contain label information. Assigning pseudo labels to unlabeled target domain data can enhance the predictive performance of transfer learning methods. The traditional adversarial domain adaptive methods may overlook the hidden label information. This leads to significant deviation and even negative transfer during domain alignment (Wu et al., 2022). The concept of the pseudo label was first originated in semi-supervised learning. The pseudo label can be divided into two categories: hard pseudo label and soft pseudo label (Lee, 2013). The hard pseudo labels directly set pseudo labels for unlabeled target domain data without considering the confidence level. For RUL prediction problems, fuzzy set theory was chosen to divide RUL labels (He, Tian, & Zuo, 2023; Wang & Tsung, 2022). He et al. (2023) and Wang and Tsung (2022) divided RUL labels into eight and three categories, respectively. The pseudo labels help calculate the marginal and conditional probability distributions in different domains. However, this method of hard pseudo label may be short of interpretability.

Another method for pseudo labeling is soft pseudo label. Soft pseudo labels can be updated during iterative training and the final category prediction can be derived by selecting the category with the highest probability (Sime et al., 2023). Soft pseudo labels are widely used in classification problems, including semantic recognition and fault diagnosis. Bae and Lim (2024) adopt a meta-pseudo-label method using a teacher/student framework with labeled and unlabeled data. The above-mentioned methods that involve soft pseudo labels are all applied in semantic recognition. Zhu, Shi, and Feng (2022) use the trained model to predict and select the classes with maximum probability as

the labels for unlabeled samples. Lu et al. (2024) use a clustering algorithm to cluster target domain samples into several classes to construct pseudo labels for target domain data. But this method may be affected by the performance of the model. Moreover, when there are incorrect classifications in the sample data, the obtained pseudo labels are also inaccurate. To solve this problem, Huo, Xu, Jiang, et al. (2023) propose a dual sample screening (DSS) method based on the information of predicted label changes. Wang et al. (2021) set pseudo labels for each subdomain to calculate the edge probability distribution and conditional probability distribution. The above-mentioned methods involving pseudo labels have achieved satisfactory results in classification problems. However, few research have explored the hidden label information in the target domain for the regression problem of RUL prediction. Few studies have proposed pseudo labeling methods with accuracy and interpretability.

Label information can reflect the health status of mechanical equipment. For the RUL prediction of mechanical equipment, assigning pseudo labels to unlabeled target domain data is essentially a transformation of health status to the RUL value. RUL value is a quantitative concept, while health status information belongs to a qualitative concept. By transforming qualitative health status into quantitative RUL value, it is possible to mine hidden label information in the target domain. Due to the randomness and ambiguity in the transformation between qualitative and quantitative concepts, the effectiveness of traditional methods is limited (Li, Liu, & Gan, 2009). Fuzzy set theory only considers the fuzzy boundaries of qualitative concept extensions, which may lead to loss of information in subsequent calculations (Liu et al., 2018). The cloud model theory (Li et al., 2009) is a new method for linguistic transformation, which is based on probability theory and fuzzy set theory. The cloud model can perfectly describe the uncertainty of qualitative concepts in the field of Failure Mode and Effects Analysis (FMEA) (Liu et al., 2023; Peng et al., 2021). It also makes the transformation between quantitative values and qualitative concepts easily and interchangeably (Wang et al., 2014). The cloud model theory based on the golden section method can effectively solve the multi-criteria group decision-making problems (Wang et al., 2015). Meanwhile, the golden section method provides a reference for determining the number of clouds. Therefore, this paper chooses to construct pseudo class labels based on the combination of golden section and cloud model theory.

The accurately prediction of the mechanical equipment RUL under cross domain conditions can effectively assist decision-makers in making reasonable usage and maintenance plans in advance. Based on the above-mentioned references, there are many urgent problems to be addressed in cross domain RUL prediction of mechanical equipment. One of the issues is to choose an appropriate method to mine the hidden label information in the target domain data. Another issue is selecting appropriate metrics to measure the transferability of source domain data. Motivated by this, Label Adversarial Domain Adaptation (LADA) network is introduced to obtain accurate RUL predictions. In the proposed method, two types of statistical metrics are adopted to characterize the similarity of the data distribution between source and target domains. Additionally, the hidden label information from target domain, termed as pseudo class label, is integrated into the model to improve the generalization ability. A feature difference module is established to reduce the discrepancy between two domain features using Maximum Mean Difference (MMD) and Kolmogorov-Smirnov (K-S) statistical. A case study on C-MAPSS dataset verifies the validity of LADA. The major contributions of this paper are as below:

(1) A new DANN framework named LADA is put forward to solve the issue of cross-domain RUL prediction considering the transferability of source domain data and the hidden label information of target domain data. The proposed method has satisfactory predictive performance and generalization ability in cross-domain RUL prediction.

(2) We design a new similarity measurement indicator to improve the transferability of source domain data. It is calculated using the desirability function based on pearson correlation coefficient (PCC) and

dynamic time warping (DTW). This indicator can guide the filtering of source domain data and improving the performance of transfer learning.

(3) In order to fully utilize the hidden label information of the target domain, a pseudo class label (PCL) module is constructed based on the cloud model and the golden section method. The generalization ability of the model is improved.

(4) A new feature difference module based on Maximum Mean Difference (MMD) and Kolmogorov–Smirnov (K–S) statistical is embedded in LADA. The feature difference module aims to further reduce the distribution differences between source domain features and target domain features.

The remainder of this paper is as follows. The proposed method is introduced in Section 2. Section 3 describes the experimental study using the proposed method to predict the RUL of the aviation engine. The prediction results of LADA, ablation experiment results and comparative experiment results have been presented in Section 3. Finally, conclusions are shown in Section 4.

2. Proposed method

In this paper, a cross-domain RUL prediction method considering the transferability of source domain data and the hidden label information of target domain data is developed. The overall framework is shown in Fig. 1. The source domain and target domain data are prepared and preprocessed before being input into the network. The network is firstly trained offline on the source and target domains to improve the prediction performance. Then the trained network is used to predict the RUL of mechanical equipment online. The offline training part of the model comprises four steps. The first step corresponds to data preprocessing, which includes source domain data filtering, data normalization, and sensor selection. In the data filtering, a similarity measurement indicator is constructed to select source domain equipment similar to the target domain data distribution. The second step involves the pre-training of the source feature extractor and RUL predictor. Then the saved network parameters are used for RUL prediction during offline training. In the third step, we construct the pseudo class label based on the cloud model and the golden section method. Finally, the corresponding network is trained using the loss function and optimizer. The second part is the online testing phase. It mainly includes two steps: data collection and processing, and RUL prediction. Data processing primarily involves normalizing the data and selecting sensors. Then, the RUL of the test set is predicted online using the network trained in the first stage.

2.1. Similarity measurement indicator

Notably, both source and target domains comprise sensor data containing degradation information from multiple equipment units. The root mean square (RMS) is widely applied to create health indicators (HI) of rotating machinery. RMS can plainly reflect degradation information in the entire lifespan of mechanical equipment (Wang & Xiang, 2021). Therefore, RMS is chosen as an index to reflect equipment degradation information. In the data filtering of source domain, the goal is to select degradation data aligns with the distribution of target domain. In order to avoid negative transfer and improve the transferability of source domain data, this paper proposes a comprehensive similarity measurement indicator. Meanwhile, in order to measure the distribution similarity of data from different domains, the indicator employs the pearson correlation coefficient (PCC) (Cohen, Huang, Chen, et al., 2009) and dynamic time warping (DTW) (Müller, 2007). PCC and DTW can respectively measure the similarity between two data points in terms of linearity (Wang et al., 2022) and non-linearity (Lu et al., 2023). It is worth noting that we only calculate the distribution similarity of corresponding sensors data from different domains. The specific steps are as follows.

Step 1: Calculate the RMS indicators of the source domain equipment and the target domain equipment.

$$S_{rms} = \begin{bmatrix} s_{11} & \cdots & s_{1q} & \cdots & s_{1Q} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ s_{i1} & \cdots & s_{iq} & \cdots & s_{iQ} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{N1} & \cdots & s_{Nq} & \cdots & s_{NQ} \end{bmatrix} \quad (1)$$

$$T_{rms} = \begin{bmatrix} t_{11} & \cdots & t_{1q} & \cdots & t_{1Q} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ t_{j1} & \cdots & t_{jq} & \cdots & t_{jQ} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t_{M1} & \cdots & t_{Mq} & \cdots & t_{MQ} \end{bmatrix} \quad (2)$$

where $i = 1, 2, \dots, N$ denotes the number of source equipment items, $j = 1, 2, \dots, M$ denotes the number of target equipment items and $q = 1, 2, \dots, Q$ denotes the number of sensors. It is worth noting that the number of sensors in the source and target domains is the same. s_{iq} denotes the RMS of the q th sensor of the i th equipment in source domain and t_{jq} represents the RMS of the q th sensor of the j th equipment in target domain.

Step 2: Calculate the PCC and DTW of the RMS matrices between the source equipment and target equipment.

$$r_{ij} = \frac{\sum_{q=1}^Q (s_{iq} - \bar{s}_{iq})(t_{jq} - \bar{t}_{jq})}{\sqrt{\sum_{q=1}^Q (s_{iq} - \bar{s}_{iq})^2} \sqrt{\sum_{q=1}^Q (t_{jq} - \bar{t}_{jq})^2}} \quad (3)$$

$$d_{ij} = \sqrt{\sum_{q=1}^Q (s_{iq} - t_{jq})^2 + \min \left\{ \begin{array}{l} \Delta(s_{iq-1}, t_{jq}) \\ \Delta(s_{iq}, t_{jq-1}) \\ \Delta(s_{iq-1}, t_{jq-1}) \end{array} \right\}} \quad (4)$$

where r_{ij} denotes the PCC between the i th source equipment and j th target equipment. d_{ij} denotes the DTW between the i th source equipment and j th target equipment. $\bar{s}_{iq}$ and $\bar{t}_{jq}$ represent the mean of s_{iq} and t_{jq} respectively.

Step 3: By calculating the mean of $r_{i1}, r_{i2}, \dots, r_{ij}, \dots, r_{iM}$, we can obtain PCC between i th equipment in source domain and all equipments in target domain. DTW was calculated in a similar manner.

$$r_i = \frac{1}{M} \sum_{j=1}^M r_{ij} \quad (5)$$

$$d_i = \frac{1}{M} \sum_{j=1}^M d_{ij} \quad (6)$$

where r_i and d_i represent the PCC and DTW between i th equipment in source domain and all equipments in target domain. r_i is between -1 and 1 ; a value closer to 1 expresses a stronger linear correlation between the two domains. For d_i , values greater than zero are expected, with a smaller d_i suggesting greater similarity between the two domains. Both PCC and DTW assess data distribution similarity between two domains on local and global scales.

Step 4: A larger PCC values is desirable, while a smaller DTW value is preferred. The desirability function method is a more reasonable and popular approach developed by Derringer and Suich (1980) to solve multi response optimization problems. Therefore, the desirability function method is employed to construct a new indicator. The individual desirability function of r_i and d_i are as follows:

$$d(r_i) = \begin{cases} 0, & r_i < L_i \\ \left(\frac{r_i - L_i}{T_{ri} - L_i} \right)^{s_i}, & L_i \leq r_i \leq T_{ri} \\ 1, & T_{ri} < r_i \end{cases} \quad (7)$$

$$d(d_i) = \begin{cases} 0, & U_i < d_i \\ \left(\frac{d_i - U_i}{T_{di} - U_i} \right)^{s_i}, & T_{di} \leq d_i \leq U_i \\ 1, & d_i < T_{di} \end{cases} \quad (8)$$

where $d(r_i)$ and $d(d_i)$ denote the individual desirability functions of two indicators. Furthermore, L_i and U_i denote the lower and upper

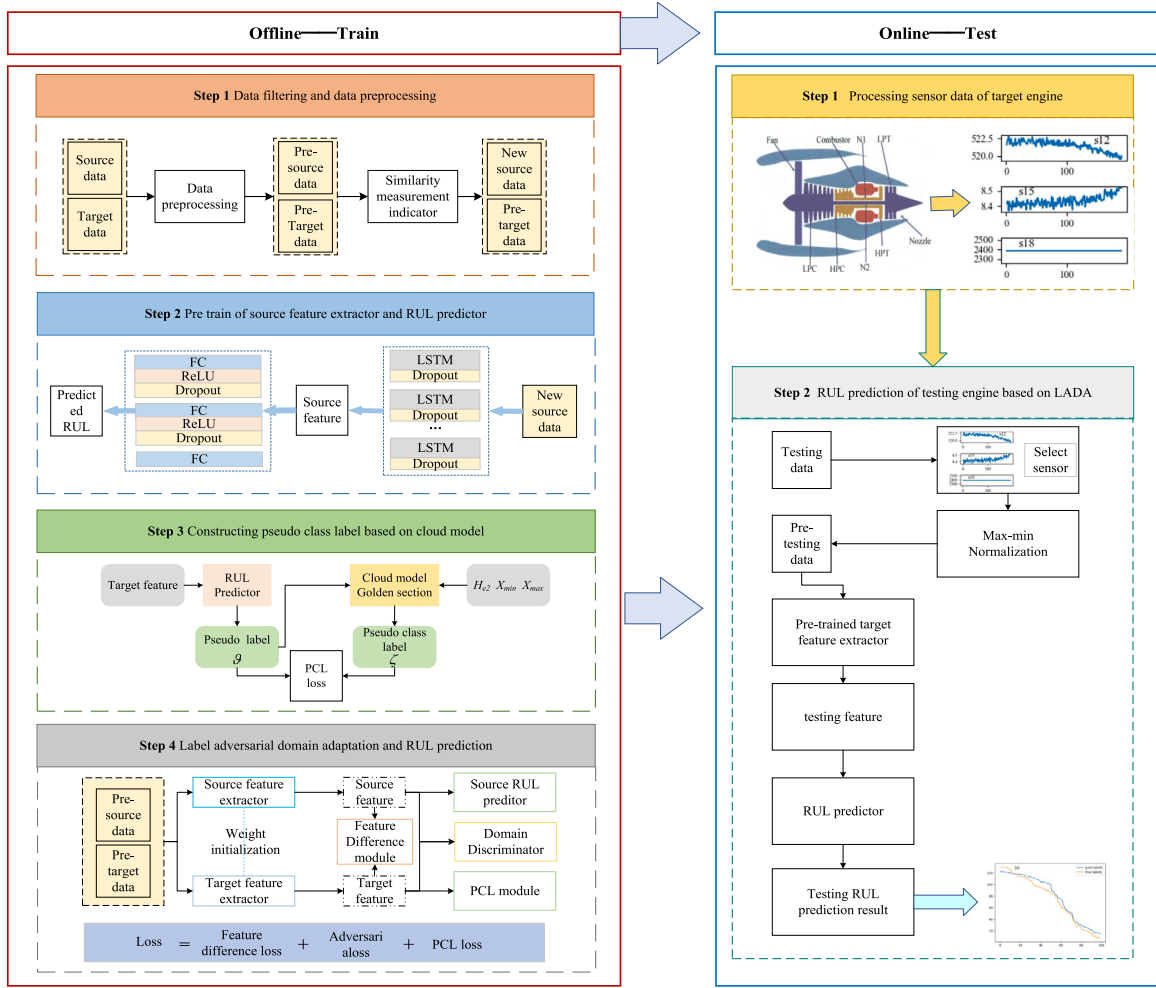


Fig. 1. The overall framework of the proposed method.

bound, respectively. T_{ri} and T_{di} denote the target value of the indicator. The index s_i determines the shape of the desirability function.

Step 5: The overall desirability function is as follows:

$$D(d(r_i), d(d_i)) = [d(r_i) \times d(d_i)]^{\frac{1}{2}} \quad (9)$$

Step 6: The mean value of the overall desirability function of all equipments in the source domain is selected as the threshold. The threshold δ is:

$$\delta = \frac{1}{N} \sum_{i=1}^N D(d(r_i), d(d_i)) \quad (10)$$

When $D(d(r_i), d(d_i)) > \delta$, the new source domain data D_{1S} can be selected based on the number of the equipment.

2.2. Pseudo class label module

In actual industrial engineering, the equipment will be repaired or replaced before it completely fails. The sensor data that can be collected is basically truncated data. Traditional research assumes that this type of data does not contain label information, they used unsupervised adaptation methods to solve cross-domain RUL prediction. Although the data in target domain were truncated, they still contain hidden label information (Zhou, 2018). In order to fully utilize the hidden label information in target domain, this paper proposes a new method for constructing pseudo class labels.

Due to the large number of labels involved in the regression problem, this paper classifies the labels to explore the hidden label information of the target domain. Yu et al. (2021) divided the degradation process of mechanical equipment into four stages: the initial period of degradation, the development period of degradation, the rapid development period of degradation and the end period of degradation. The health status is added to these four degradation stages, forming a five stages classification method. The entire lifecycle of mechanical equipment is divided into five stages: health state (HS), early degradation state (EDS), slow degradation state (SDS), fast degradation state (FDS) and near failure state (NFS). Subsequently, the cloud model and golden section are used to classify the source domain labels into pseudo class labels. The specific steps for constructing pseudo class labels are as follows.

Step 1: We assume that the effective domain $U = [X_{\min}, X_{\max}]$. Based on the above five stages, we can obtain a linguistic term set $Z = \{z_0, z_1, \dots, z_g\}$. g is the number of semantic labels and g denotes an even integer. In this paper, $g = 4$, the linguistic label set Z is as follows:

$$Z = \left\{ \begin{array}{l} z_0 = \text{Near Failure State (NFS)} \\ z_1 = \text{Fast Degradation State (FDS)} \\ z_2 = \text{Slow Degradation State (SDS)} \\ z_3 = \text{Early Degradation State (EDS)} \\ z_4 = \text{Health State (HS)} \end{array} \right\} \quad (11)$$

Step 2: Based on the linguistic labels mentioned above, the basic cloud corresponding to the linguistic label can be obtained. Each basic

cloud contains three numerical features, namely (E_{xg}, E_{ng}, H_{eg}) .

$$\begin{cases} \hat{y}_0 = (E_{x0}, E_{n0}, H_{e0}) \\ \hat{y}_1 = (E_{x1}, E_{n1}, H_{e1}) \\ \hat{y}_2 = (E_{x2}, E_{n2}, H_{e2}) \\ \hat{y}_3 = (E_{x3}, E_{n3}, H_{e3}) \\ \hat{y}_4 = (E_{x4}, E_{n4}, H_{e4}) \end{cases} \quad (12)$$

where $\hat{y}_g$ represents the basic cloud generator. E_{xg} is the central value of the domain, so it can represent the label value. E_{ng} can reflect the fuzziness of the domain, while H_{eg} reflects the random changes in the dispersion and membership of cloud droplets.

Step 3: Represent linguistic labels based on the golden section method. 0.618 is set as the multiple between the numerical features of adjacent linguistic labels. The closer to the central region of the domain $[X_{\min}, X_{\max}]$, the smaller the values of E_{ng} and H_{eg} in the evaluation level. Based on golden section, the expression of three numerical features of five basic clouds can be obtained.

$$\begin{cases} \hat{y}_0 = (E_{x0}, E_{n0}, H_{e0}) = (X_{\min} + 3E_{n0}, \frac{E_{n1}}{0.618}, \frac{H_{e1}}{0.618}) \\ \hat{y}_1 = (E_{x1}, E_{n1}, H_{e1}) = (E_{x2} - 0.382(E_{x2} - E_{x0}), \frac{E_{n2}}{0.618}, \frac{H_{e2}}{0.618}) \\ \hat{y}_2 = (E_{x2}, E_{n2}, H_{e2}) = (\frac{X_{\min} + X_{\max}}{2}, \frac{0.382(X_{\max} - X_{\min})}{3(g+2)}, H_{e2}) \\ \hat{y}_3 = (E_{x3}, E_{n3}, H_{e3}) = (E_{x2} + 0.382(E_{x4} - E_{x2}), \frac{E_{n2}}{0.618}, \frac{H_{e2}}{0.618}) \\ \hat{y}_4 = (E_{x4}, E_{n4}, H_{e4}) = (X_{\max} - 3E_{n4}, \frac{E_{n3}}{0.618}, \frac{H_{e3}}{0.618}) \end{cases} \quad (13)$$

Step 4: For the above equation, determining the values of the three unknown parameters can obtain the values of the remaining thirteen parameters. These three unknown parameters are $X_{\max}$, $X_{\min}$ and H_{e2} . In this paper, $X_{\max}$ and $X_{\min}$ represent the maximum and minimum values of RUL labels, respectively. We assume $X_{\max} = 0$ and $X_{\min} = 130$. The value of H_{e2} needs to be determined in subsequent experiments (Saxena, Goebel, Simon, others, 2008). Based on this, we can obtain five cloud generators.

$$\begin{cases} \hat{y}_0 = (E_{x0}, E_{n0}, H_{e0}) = (21.672, 7.224, \frac{H_{e2}}{0.618^2}) \\ \hat{y}_1 = (E_{x1}, E_{n1}, H_{e1}) = (48.449, 4.464, \frac{H_{e2}}{0.618}) \\ \hat{y}_2 = (E_{x2}, E_{n2}, H_{e2}) = (65, 2.758, H_{e2}) \\ \hat{y}_3 = (E_{x3}, E_{n3}, H_{e3}) = (85.551, 4.464, \frac{H_{e2}}{0.618}) \\ \hat{y}_4 = (E_{x4}, E_{n4}, H_{e4}) = (108.328, 7.224, \frac{H_{e2}}{0.618^2}) \end{cases} \quad (14)$$

Step 5: For a given pseudo label, the membership degree v_g of this pseudo label belonging to each linguistic label can be obtained. And $\sum_g v_g = 1$. The g th cloud corresponding to $\max v_g$ is the linguistic label to which it belongs. Then assign E_{xg} as the pseudo class label for this label.

If a target domain sample has a predicted pseudo label of 40 during the iteration period. It can be obtained by passing through the five cloud models of the PCL module with a corresponding pseudo class label of E_{x2} . Additionally, each pseudo label only corresponds to only one pseudo class label. The target domain samples are extracted by target feature extractor to obtain target feature $f_T = F_T(X_T)$. The target feature is input into the PCL module to obtain the target domain RUL prediction value, which is the pseudo label (PL) of the target domain. The pseudo label is fed into the cloud model to obtain the pseudo class label (PCL) of this pseudo label. The flowchart of the PCL module is shown in Fig. 2. The loss of the PCL module is expressed as follow:

$$\mathcal{L}_{PCL} = \frac{1}{n_T} \sum_{l=1}^{n_T} (\theta_l - \zeta_l)^2 \quad (15)$$

where $l = 1, 2, \dots, n_T$ represents the number of samples in target domain, θ_l and ζ_l represent the pseudo labels and pseudo class labels of target domain samples, respectively.

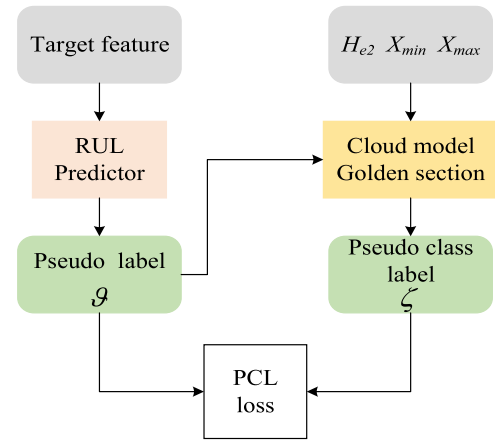


Fig. 2. The flowchart of PCL loss.

2.3. Feature difference module

In order to solve the issue of data distribution differences caused by changes in operating conditions, this paper conducts a feature difference module (FDM) to minimize the discrepancy between features from different domains. On this basis, the adversarial training of feature extractors and domain discriminators is retained. Firstly, the pre-trained source model weights are used to initialize F_T . Then, the features extracted by F_T and F_S are used to improve the discrimination ability of domain discriminator D . This poses a challenge for D to distinguish between the target and source features. The loss of the feature difference module is used to reduce the distribution difference between the two domain features. The adversarial training technology between D and F_T can be represented as follows:

$$\min_{F_T} \max_D \mathcal{L}_{adv} = E_{X_S \sim P_S} [\log D(F_S(X_S))] + E_{X_T \sim P_T} [\log (1 - D(F_T(X_T)))] \quad (16)$$

where X_S and X_T represents the source and target samples, respectively. The parameters of F_T and D are updated by minimizing the adversarial loss $\mathcal{L}_{adv}$.

The feature difference module integrates the maximum mean difference (MMD) and Kolmogorov–Smirnov (K–S) statistical to measure the differences between two domain features. MMD measures the data similarity from the perspective of mean differences, whereas K–S measures the similarity from the perspective of maximum differences. The MMD losses are expressed as:

$$\mathcal{L}_{MMD} = MMD(f_S, f_T) = \left\| \frac{1}{n_S} \sum_k \phi(f_S^k) - \frac{1}{n_T} \sum_l \phi(f_T^l) \right\|_{\mathcal{H}} \quad (17)$$

where $\|\cdot\|_{\mathcal{H}}$ denotes the regenerated kernel Hilbert space (RKHS), and $\phi(\cdot)$ denotes mapping source features and target features to RKHS.

The K–S statistical test values between the source feature distribution P_{f_S} and target features distribution P_{f_T} are as follows:

$$N_e = \sqrt{\frac{n_S \times n_T}{n_S + n_T}} \quad (18)$$

$$K = \sup_{-\infty < x < \infty} |P_{f_S}(x_{n_S}) - P_{f_T}(x_{n_T})| \quad (19)$$

where n_S and n_T represent the lengths of source domain features and target domain features. The probability of similarity $\mathcal{P}$ can be determined by the reliability distribution function Q_{ks} :

$$\lambda = \left(\sqrt{N_e} + 0.12 + \frac{0.11}{\sqrt{N_e}} \right) \times K \quad (20)$$

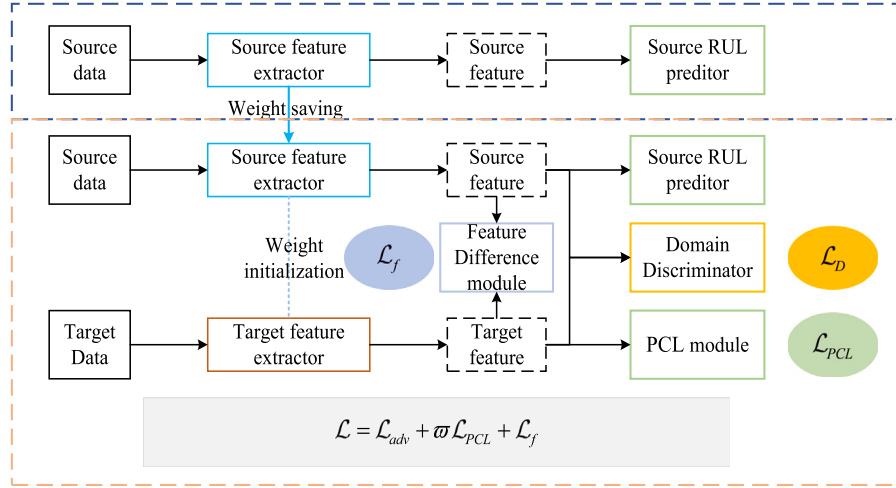


Fig. 3. Framework of the proposed method.

$$P(K) = Q_{ks}(\lambda) = 2 \sum_{i=1}^{+\infty} (-1)^{i+1} \exp^{-2i^2 \lambda^2} \quad (21)$$

Comparing the P value with a given confidence ξ . if $P > \xi$, it can be considered that there is no significant difference between two data. During the training process, it is expected that P has a gradient optimization in the direction of increasing. Therefore, the loss of the feature difference module can be expressed as:

$$\mathcal{L}_f = \mathcal{L}_{MMD} - P(K) \quad (22)$$

The final trained target feature extractor can extract target features that how minimal deviation from the source domain features.

2.4. Label adversarial domain adaptation

The Label Adversarial Domain Adaptation (LADA) network proposed in this paper mainly includes five parts: feature extractor, RUL predictor, domain discriminator, pseudo class label (PCL) module and feature difference module (FDM). Fig. 3 shows the proposed LADA.

Firstly, the source domain feature extractor F_S and RUL predictor are pre-trained using labeled source domain data, and their parameters are saved. The target domain feature extractor F_T is initialized using the weights of the pre-trained F_S . The features of source and target domains can be extracted. The features of both domains are put into the domain discriminator D to recognize the domain from which the features originated. To minimize the difference of domain distribution, we add the MMD loss and K-S statistical test as the feature difference loss. At the same time, the target features are put into the PCL module for pseudo class label loss calculation, in order to explore the hidden label information from target data.

In this paper, adversarial training loss, pseudo class label loss, and feature difference loss are used to jointly optimize end-to-end network models. The overall loss function is expressed as follows:

$$\min_{F_T} \max_D (D, F_T, \Theta) = \mathcal{L}_{adv} + w\mathcal{L}_{PCL} + \mathcal{L}_f \quad (23)$$

where $\mathcal{L}_{adv}$ is the adversarial loss, $\mathcal{L}_{PCL}$ is the pseudo class label loss and $\mathcal{L}_f$ is the feature difference loss. This paper argues that the roles of feature extractors, domain discriminators, and feature difference modules are equally important. The loss weights of $\mathcal{L}_{adv}$ and $\mathcal{L}_f$ are both set to 1. w is a weight parameter to control the learning of hidden label information in target domain features.

The flow chart of the prediction procedure is presented in Fig. 4, which contains three parts. Firstly, the similarity measurement indicator is used to filter the source domain data and the filtered source

domain data is used to pre-train the network. The training and testing samples can be obtained via data pre-preprocessing. Next, the network architecture is determined to solve cross-domain RUL prediction. Both source data with labels and target data without labels are sent into the network to start the learning process. Deep features of the sensor data are extracted through the weighted initialized network. The target domain data is first used to construct pseudo class labels. Then, the adversarial loss, feature difference loss, and pseudo class label loss of the network are calculated based on the deep representation of the domain. When the number of training periods reaches the goal of training times, the training of the network is finished. After the training process, the test samples of target domain are fed back to the proposed network to get the prediction results of the test RUL.

Algorithm 1 describes the formal process of the proposed LADA approach. The domain discriminator D aims to distinguish the features of two domains. In comparison, the feature difference module aims to narrow the distribution discrepancy between the features of two domains. The negative impact of variable operating conditions can be reduced through such adversarial mechanism.

Algorithm 1: Pseudo Class Label Adversarial Domain Adaptation

Data: Source domain data: $D_S = \{X_S^k, Y_S^k\}_{k=1}^{n_S}$

Target domain data: $D_T = \{X_T^l\}_{l=1}^{n_T}$

$F_S \leftarrow$ Trained source feature extractor

$F_T \leftarrow$ Initialize with F_S parameters

$D \leftarrow$ Domain discriminator

1 for number of iterations do

2 Sample minibatch of source samples $X_S \sim P_S$;

3 Sample minibatch of target samples $X_T \sim P_T$;

4 Extract source features: $f_S = F_S(X_S)$;

5 Extract target features: $f_T = F_T(X_T)$;

6 Feed f_S and f_T to D ;

7 Compute adversarial loss $\mathcal{L}_{adv}$ by Eq. (16);

8 Update D by $\mathcal{L}_{adv}$;

9 Compute PCL loss by Eq. (15);

10 Compute feature difference loss $\mathcal{L}_f$ by Eq. (22);

11 Update F_T by $\mathcal{L} = \mathcal{L}_{adv} + w\mathcal{L}_{PCL} + \mathcal{L}_f$

12 end

Result: Trained target feature extractor: F_T

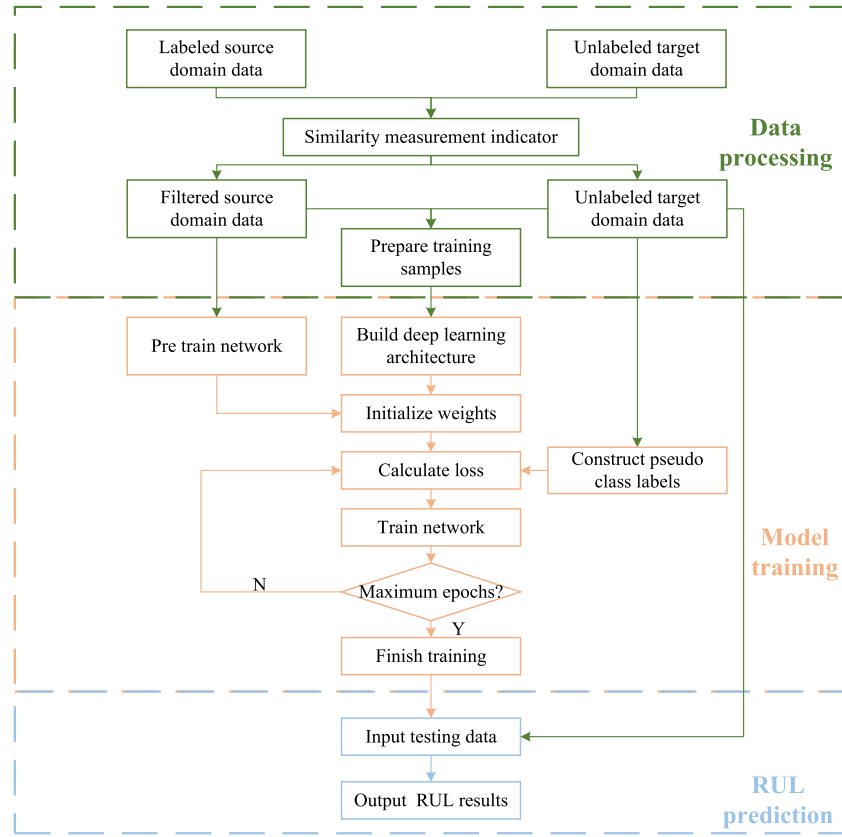


Fig. 4. Flow chart of the proposed cross-domain RUL prediction method.

Table 1
C-MAPSS dataset.

Data Subset	FD001	FD002	FD003	FD004
Engines: Training	100	260	100	249
Engines: Testing	100	259	100	248
Training: sample size	17 731	48 558	21 220	56 815
Operating Condition (OC)	1	6	1	6
Fault Modes (FM)	1	1	2	2

3. Experimental study

3.1. Data description and preparation

To evaluate the performance of LADA, this paper uses the Commercial Modular Aero Propulsion System Simulation (C-MAPSS) benchmark dataset (Saxena et al., 2008). The data set includes four data subsets: FD001, FD002, FD003 and FD004. Each data subset is different in terms of failure mode, working condition, lifespan and engines number. The parameters of the subset data are shown in Table 1. Each engine has different degrees of initial wear, which is considered to be in a healthy state at the beginning of the cycle. With the increase of time, the performance of each unit gradually decreases until it reaches the failure state and loses its functionality. The training set includes the complete cycle from operation to broken, and the test set includes the truncated trajectory before broken.

In practical application scenarios, massive sensors are installed on the mechanical equipment to monitor its running state. The data collected by the sensor can reflect the health status of the equipment. However, not all sensor data contain degradation information of the equipment. Some sensors are almost constant during the whole degradation process. Most researchers believe that such sensor data does not contain degradation information that is helpful for RUL prediction,

and will prevent the network from modeling the degradation procedure correctly. To improve RUL prediction accuracy, this paper chooses the sensor that is useful for RUL prediction. Sensors with unchanged data are removed from the original data. Therefore, this paper only retains 14 sensors with degradation trends during data processing (Xu et al., 2022).

To eliminate the dimensional influence between the features of each dimension, data normalization processing is used to solve the comparability between the features of each dimension. Therefore, this paper selects the Max-min Normalization method to preprocess the data:

$$x_{norm} = \frac{x_i - x_{min}}{x_{max} - x_{min}} \quad (24)$$

where x_{max} and x_{min} denote the maximum and minimum of the samples respectively.

3.2. Experimental settings

The structure of the network modules included in LADA is shown in Fig. 5. The structure of the RUL predictor is the same as that of the PCL module. To conduct the cross-domain RUL prediction experiments, this paper divides the data into source and target domains. There are twelve transfer tasks as shown in Table 2. Different transfer tasks have different learning rates. By trying different parameter values and comparing the performance and convergence under different tasks, the optimal combination of hyperparameters will be found. We select Adam as the optimizer to set different learning rates. The exponential decay rates used to calculate the first and second moments are set as (0.5, 0.9) in each task. The batch size is set to 256 and the sliding time window size is 30.

Fig. 6 shows the basic cloud under different parameters. Among them, H_{e2} is set to 0.01 (Wu, Zhen, & Zhang, 2020), 0.02 (Liu et al.,

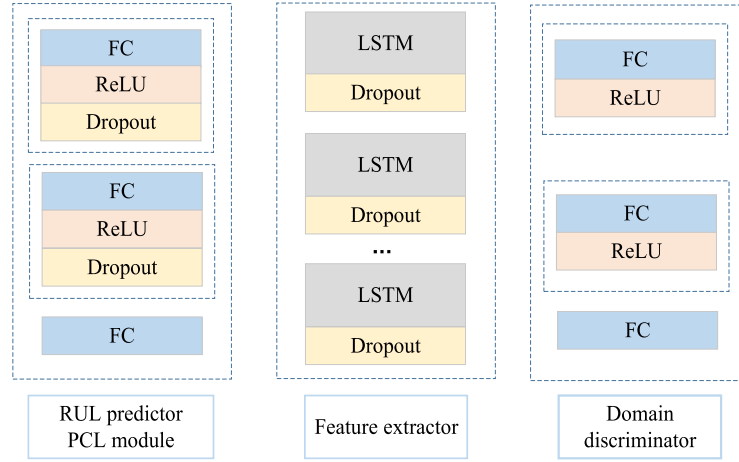


Fig. 5. The structure of the network modules of LADA.

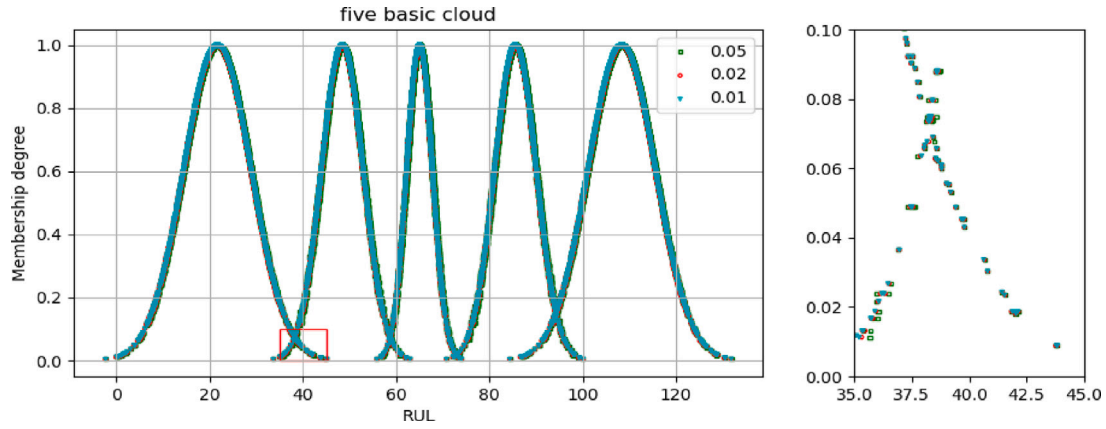


Fig. 6. Parameter analysis of cloud models.

Table 2

Twelve transfer tasks.

Source domain	Target domain	Source domain	Target domain
FD001	FD002	FD003	FD001
FD001	FD003	FD003	FD002
FD001	FD004	FD003	FD004
FD002	FD001	FD004	FD001
FD002	FD003	FD004	FD002
FD002	FD004	FD004	FD003

2018) and 0.05 (Zhao, Li, Wang, & Yuan, 2020). The figure on the right shows a partial enlarged view of the basic cloud with different parameters. It can be seen that there is no difference in the shape of clouds under different parameters. At the end of the cloud, the degree of dispersion of cloud droplets varies. The larger the H_{e2} , the more dispersed the cloud droplets become. Due to the small parameter values, the difference between them is also minimal. Therefore, this article sets H_{e2} to 0.02.

The sliding time window method is used to process the input data. The minimum number of training data cycle is 31. The sliding time window size is set as 30 and the time step is set as 1. Li et al. (2023) proved that setting the RUL value to 130 is reasonable and can reduce the number of labels. Thus, the RUL label value is assumed as a piecewise linear function, and the maximum value of RUL is set as 130. The piecewise linear function is shown in Fig. 7. When the RUL value is greater than 130, the equipment is in a healthy state.

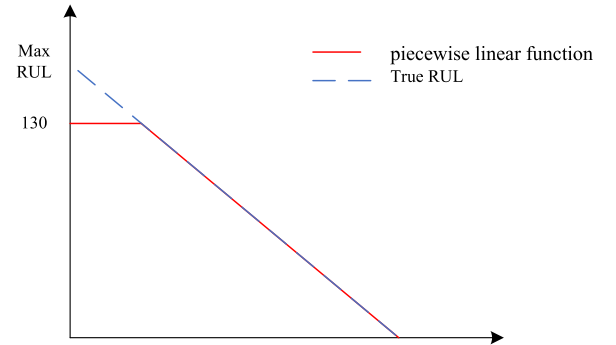


Fig. 7. The piecewise linear function.

To quantify the prediction performance of the model, two evaluation indicators are employed: the Root Mean Square Error (RMSE) and Score. The expressions of RMSE and Score are as follow:

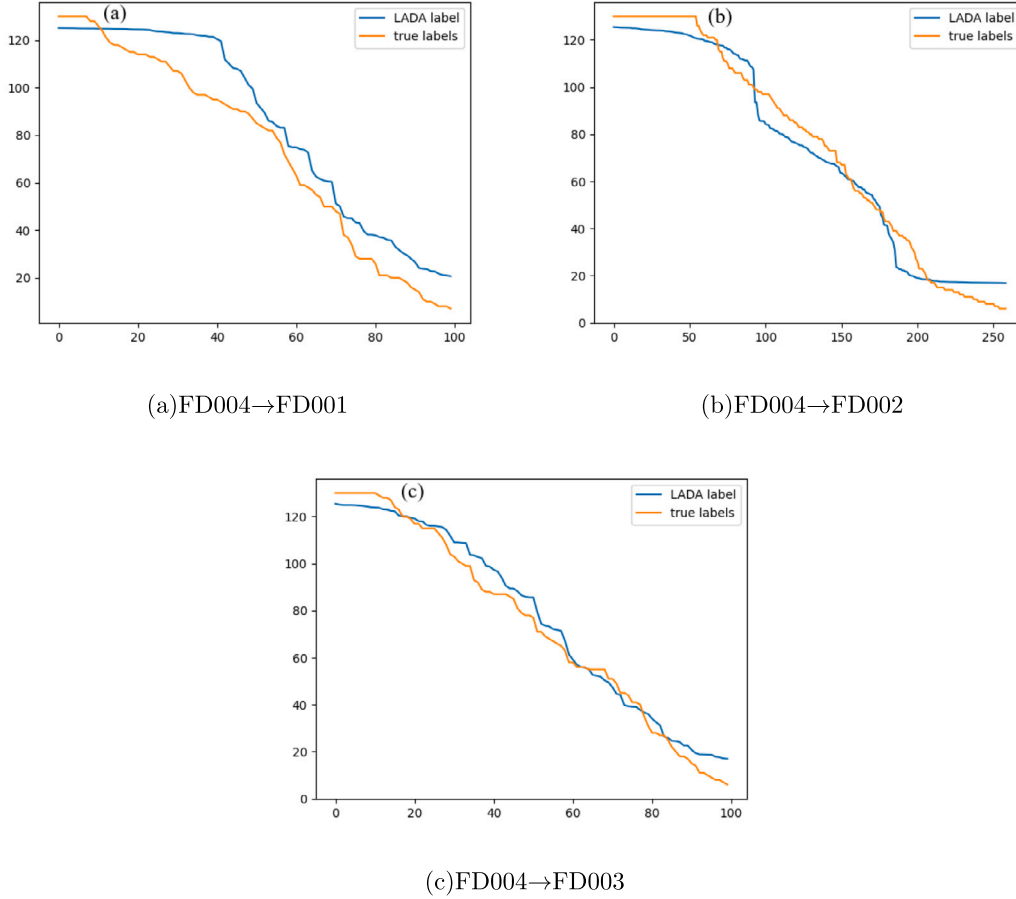
$$RMSE = \sqrt{\frac{1}{n_T} \sum_{l=1}^{n_T} (\hat{y}_l - y_l)^2} \quad (25)$$

$$Score = \begin{cases} \frac{1}{n_T} \sum_{l=1}^{n_T} e^{\frac{\hat{y}_l - y_l}{13} - 1}, & \hat{y}_l < y_l \\ \frac{1}{n_T} \sum_{l=1}^{n_T} e^{\frac{\hat{y}_l - y_l}{10} - 1}, & \hat{y}_l > y_l \end{cases} \quad (26)$$

Table 3

The performance of LADA under twelve transfer tasks.

Transfer tasks	RMSE	Score	Transfer tasks	RMSE	Score
FD001→FD002	19.74	3949.27	FD003→FD001	14.76	528.34
FD001→FD003	29.92	4013.34	FD003→FD002	22.36	14831.46
FD001→FD004	31.17	6860.69	FD003→FD004	20.56	2944.50
FD002→FD001	13.74	345.03	FD004→FD001	21.76	2395.77
FD002→FD003	21.09	1138.53	FD004→FD002	17.30	3596.47
FD002→FD004	19.29	5331.18	FD004→FD003	13.89	597.53

**Fig. 8.** RUL fitting results with source domain FD004.

where $\hat{y}_i$ denotes the predicted value of network and y_i denotes the true label value.

3.3. RUL prediction results and discussion

Table 3 shows the performance of LADA under twelve transfer tasks. From Table 3, the proposed method has satisfactory performance. The RMSE for most transfer tasks remains around 20. For FD002→FD001 and FD004→FD003, lower RMSE was achieved than other transfer tasks. For the transfer tasks when FD002 and FD004 are chosen as source domains, LADA achieved mean RMSE values of 18.04 and 17.65 respectively. This indicates that LADA can achieve excellent prediction results when the source domain data contains more fault modes and operating conditions. When FD001 and FD003 are respectively chosen as source domains, LADA achieved results of 26.94 and 19.22. LADA has poor predictive performance when used FD001 as the source domain. Due to the additional fault mode, the performance of FD003 as the source domain has been improved. This may indicate that the failure mode of the equipment can provide more degradation information than operating conditions.

Fig. 8 shows the RUL fitting results of the transfer task using FD004 as source domains. The curve of the true values shown in Fig. 8 has been processed in descending order. LADA delivers satisfactory results under cross-domain RUL prediction. The predicted results are close to the real RUL values. It is worth noting that when the RUL value of the engine is small, the predicted result is larger than the true value. That is, there is a deviation between the predicted value and the true value at the end of the fitting line. This deviation is acceptable because the engine's lifespan will be replaced or repaired before reaching to a very small value. Based on this deviation, engineers can set a threshold. This threshold can guide engineers to revise maintenance scheduling plans in a timely manner.

3.4. Ablation experiments results and discussion

To verify the effectiveness of the LADA method, four ablation experiments were conducted in this section.

3.4.1. Data filtering results and discussion

The source domain engine data is selected to form the new source data when the similarity measurement indicator is greater than the

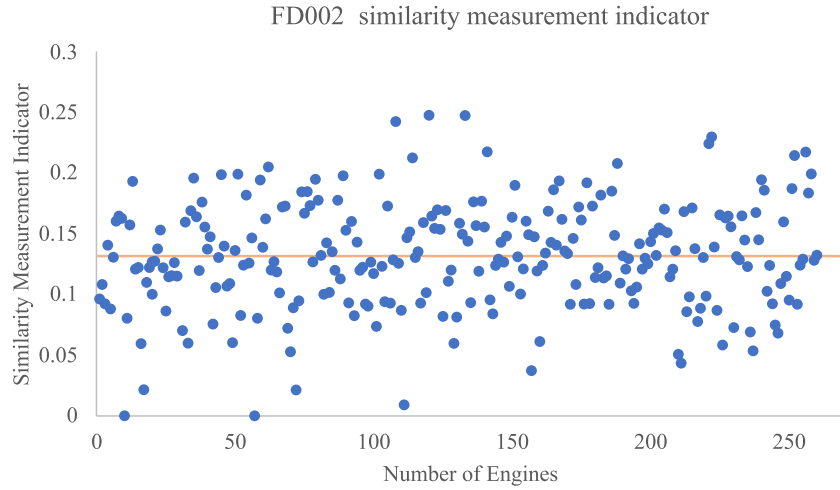


Fig. 9. Data Filtering.

Table 4
The engine similarity threshold.

Transfer task	threshold	Transfer task	threshold
FD001→FD002	0.00630	FD003→FD001	0.43531
FD001→FD003	0.81207	FD003→FD002	0.01797
FD001→FD004	0.00615	FD003→FD004	0.01797
FD002→FD001	0.13153	FD004→FD001	0.11837
FD002→FD003	0.13180	FD004→FD002	0.64228
FD002→FD004	0.67781	FD004→FD003	0.11859

corresponding threshold. The thresholds of the similarity measurement indicators of the twelve transfer tasks are obtained based on Subsection 2.1, which can be seen in Table 4. The similarity measurement indicator value of FD002→FD001 in the source domain is shown as Fig. 9. The orange line indicates the threshold and the engine corresponding to the similarity measurement indicator above the line is selected to form the new source domain data.

The filtered source data and target data are input into the LADA to predict the RUL of mechanical equipment. Table 5 shows the comparison results of the predictive performance of different networks. “without” implies that the test results obtained without data filtering. From Table 5, it is evident that data filtering can improve transfer learning performance. The RMSE and Score of the predicted results with data filtering are significantly lower than those without data filtering. Based on the average results when compared with the case without data filtering, LADA reduced RMSE by over 20% and Score is improved by more than 50%. Among them, the domain shift between FD001 and FD004 is the most obvious. However, the improvement effect is best under the FD004→FD001 transfer task. This indicates that a larger disparity in the data distribution between source and target domains amplifier the importance of data filtering. Consequently, the improvement of transfer effect is most evident in FD002→FD003 and FD004→FD001. Data filtering can significantly reduce the negative transfer impact caused by data distribution differences. The similarity measurement indicator proposed in this paper can measure the similarity between the two data.

3.4.2. Ablation experiments

In this section, the ablation experiments are described to verify the contribution of the domain discriminator, feature difference module and PCL module. The experimental settings are as follows: “source-only” implies not using the TL strategy; “without-FDM” represents removing the feature difference module in the LADA method proposed in this paper, “without-PCL” implies that the test results obtained by removing the PCL module from the LADA method. The RMSE and

Score results of the ablation experiment are shown in Tables 6 and 7, respectively. From the ablation experiment results in the table, it can be seen that the LADA method has a more significant improvement effect than the “source-only” and “without-FDM” methods. Especially when the distribution differences between the two domains are relatively large, such as FD002→FD003 and FD004→FD001. This indicates that transfer learning can solve RUL prediction under multiple working conditions. Moreover, the feature difference module can also improve the performance of the network by reducing the difference in feature distribution between the two domains. The RMSE and Score of LADA decreased by 23.33% and 93.80% respectively compared to “without-PCL”. The comparison results indicate that the PCL module has further improved the performance of the network on the basis of the domain discriminator and feature difference module. This also proves that the truncated data in the target domain does contain certain label information. Exploring the hidden label information of the target domain can improve the predictive performance of transfer learning networks.

Fig. 10 shows the RUL fitting results of the ablation experiment using FD001 as the source domain. From Fig. 10b and Fig. 10c, it can be seen that the improvement effect of LADA is more significant when there is a greater difference in data distribution. The performance of LADA method is superior to the other three ablation experiments among the three transfer tasks. This indicates that mining the hidden label information in the target domain and reducing the differences between features can significantly improve the predictive performance of TL.

In addition, to verify the effectiveness of the PCL module, we visualized the PCL loss. Fig. 11 shows the PCL loss using FD002 as source domain. It can be seen that the PCL loss shows a significant downward trend and converges when the training frequency is around 20 times. The losses under all four transfer tasks decrease from around 500 to around 30. This also indicates the effectiveness of the pseudo class label method proposed in this paper.

3.4.3. Comparative experiments

To evaluate the RUL prediction performance of LADA in cross-domain scenarios, we select six advanced transfer learning methods for comparison, including:

(1) Deep representation regularization (DRR) (Zhang et al., 2021): a transfer learning method for remaining useful life predictions using deep representation regularization.

(2) Deep domain adaptation (DDA) (da Costa et al., 2020): DDA combines the domain antagonistic neural network with LSTM for RUL prediction task.

Table 5

The comparison results of the predictive performance.

Metrics	RMSE			Score		
Transfer task	without	LADA	rel. Delta	without	LADA	rel. Delta
FD001→FD002	21.72	19.74	−9.11%	8999.51	3949.27	−56.11%
FD001→FD003	40.29	29.92	−25.73%	10595.76	4013.34	−27.91%
FD001→FD004	33.42	31.17	−6.735%	14419.54	6860.69	−95.42%
FD002→FD001	14.42	13.74	−4.72%	467.40	345.03	−26.18%
FD002→FD003	34.79	21.09	−39.38%	5711.35	1138.53	−80.07%
FD002→FD004	28.23	19.29	−31.67%	49777.53	5331.18	−89.29%
FD003→FD001	16.41	14.76	−10.05%	2910.10	528.34	−81.84%
FD003→FD002	25.93	22.36	−13.77%	34379.61	14831.46	−56.86%
FD003→FD004	23.18	20.56	−11.30%	14503.95	2944.50	−79.7%
FD004→FD001	35.76	21.76	−39.15%	105400.78	2395.77	−96.34%
FD004→FD002	18.37	17.30	−5.82%	4546.23	3596.47	−20.89%
FD004→FD003	17.06	13.89	−18.58%	1441.56	597.53	−58.55%

Table 6

The RMSE of the ablation experiments.

Transfer task	Source-only	Without-FDM	Without-PCL	LADA
FD001→FD002	24.80	21.22	20.49	19.74
FD001→FD003	46.79	39.79	38.89	29.92
FD001→FD004	37.80	32.52	33.80	31.17
FD002→FD001	15.29	14.63	14.61	13.74
FD002→FD003	35.46	32.13	28.78	21.09
FD002→FD004	37.87	34.18	35.50	19.29
FD003→FD001	15.80	15.34	15.03	14.76
FD003→FD002	46.11	29.35	30.78	28.13
FD003→FD004	31.44	22.92	28.44	20.56
FD004→FD001	37.9	34.78	33.43	21.76
FD004→FD002	37.75	32.29	33.02	17.30
FD004→FD003	19.47	14.96	15.11	13.89
Mean	31.85	27.01	27.32	20.95

Table 7

The Score of the ablation experiments.

Transfer task	Source-only	Without-FDM	Without-PCL	LADA
FD001→FD002	9128.80	7537.92	5893.80	3949.27
FD001→FD003	968784.99	13249.17	10269.73	4013.34
FD001→FD004	23923.52	14207.49	15060.67	6860.69
FD002→FD001	543	465.57	483.18	345.03
FD002→FD003	5339.31	6114.83	4867.87	1138.53
FD002→FD004	757830.90	10031.30	13888.24	5331.18
FD003→FD001	731.25	671.69	566.16	528.34
FD003→FD002	72405.04	56380.21	64459.80	14831.46
FD003→FD004	40772.46	4030.05	9059.65	2944.50
FD004→FD001	99596.66	59541.68	65967.24	2395.77
FD004→FD002	62344.87	54531.15	59056.72	3596.47
FD004→FD003	2469.73	847.63	897.4	597.53
Mean	170052.96	18967.39	20872.54	3028.56

(3) Correlation alignment (CORAL) (Sun et al., 2017): an approach to align the distribution by minimizing the covariance shift in different domains.

(4) Wasserstein distance guided representation learning (WDGRL) (Shen et al., 2018): a novel method to learn domain invariant feature.

(5) Deep domain confusion (DDC) (Tzeng et al., 2014): a new CNN to learn domain invariant features.

(6) Contrastive adversarial domain adaptation (CADA) (Ragab et al., 2020): an approach with a contrastive loss based on adversarial domain adaptation network to consider target-specific information.

(7) Weighted adversarial loss DANN (WAL-DANN) (Wu et al., 2022): WAL utilizes the ground-truth labels of the source domain and the pseudo-labels of the target domain to calculate weight and obtain the WAL.

(8) Adversarial discriminative domain adaptation (ADDA) (Tzeng et al., 2017): ADDA uses a GAN-based loss to determine the target characteristics matched with the source characteristics.

Table 8 shows the comparison results of the RMSE. Table 9 shows the comparison results of the Score. From Tables 8 and 9, the proposed LADA outperforms the first five methods in all transfer tasks. This

indicates that the proposed method is superior to other baseline methods. When using FD002 as source domains, the performance of LADA has better performance than the WAL-DANN, DRR and CADA. Under the FD004→FD001 transfer task, the performance of LADA is close to CADA. Under other transfer tasks with FD004 as the source domain, LADA performs the best. Overall, LADA performs satisfactory when using FD002 and FD004 as the source domain. When using FD003 as the source domain, LADA achieved the best prediction accuracy in both FD003→FD001 and FD003→FD004. The performance of LADA has slightly decreased in transfer tasks in FD003→FD002. When FD001 as the source domain, LADA's performance is not satisfactory. As a subset of data containing the least number of failure modes and operating conditions, FD001 poses a challenge in transferring to complex target domain data. From the mean of the RMSE and score, the LADA method is significantly superior to the other methods. This also proves the effectiveness of the proposed similarity measurement indicator and pseudo class label method.

FD001 only contains one fault mode, while FD003 contains two fault modes. Both FD001 and FD003 only contain one operating condition. The transfer tasks when using FD003 as the source domain

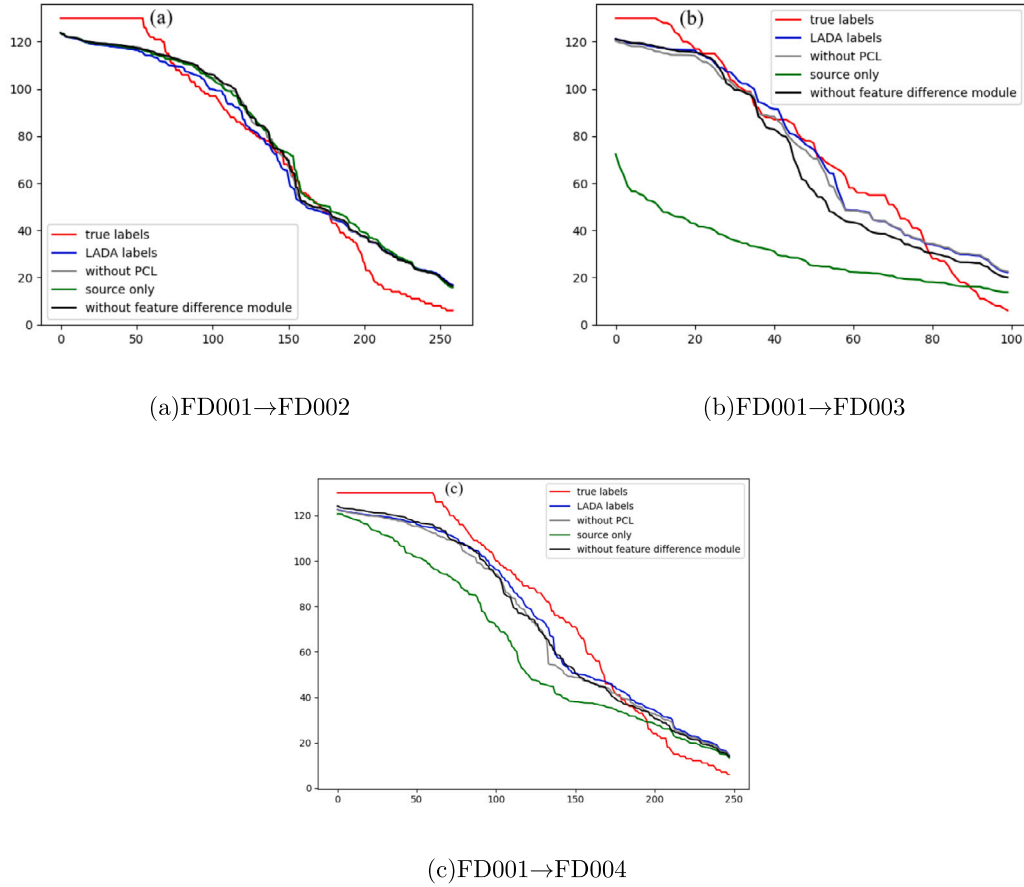


Fig. 10. RUL fitting results of ablation experiment with source domain FD001.

Table 8

Comparison results of RMSE.

RMSE	CORAL	WDGRL	DDC	ADDA	DDA	WAL-DANN	DRR	CADA	LADA
FD001→FD002	22.85	21.46	44.05	31.26	24.08	20.95	43.4	19.52	19.74
FD001→FD003	44.21	71.7	39.62	57.09	43.08	25.64	24.5	39.58	29.92
FD001→FD004	50.03	57.24	44.35	56.66	45.7	24.21	45.1	31.23	31.17
FD002→FD001	24.43	15.24	46.96	19.73	23.91	21.49	34.6	13.88	13.74
FD002→FD003	42.66	41.45	39.87	37.22	47.26	32.53	43.2	33.53	21.09
FD002→FD004	52.12	37.62	43.99	37.64	45.17	25.54	33.1	33.71	19.29
FD003→FD001	40.33	36.05	39.95	40.41	27.15	21.26	18.3	19.54	14.76
FD003→FD002	56.67	40.11	44.07	42.53	30.42	22.27	48.8	19.33	22.36
FD003→FD004	38.16	29.98	47.46	31.88	31.82	24.52	52.5	20.61	20.56
FD004→FD001	51.44	42.01	41.55	37.81	32.37	24.97	35.3	20.1	21.76
FD004→FD002	31.61	35.88	43.99	36.67	27.54	23.32	24.6	18.5	17.30
FD004→FD003	30.44	18.18	44.47	23.59	23.31	26.01	38.7	14.49	13.89
Mean	40.41	37.24	43.36	37.71	33.48	24.39	36.84	23.67	21.67

Table 9

Comparison results of Score.

Score	CORAL	WDGRL	DDC	ADDA	DDA	WAL-DANN	CADA	LADA
FD001→FD002	2789	33 160	5958	4865	2684	2707	2122	3949.27
FD001→FD003	56 991	15 936	288 061	32 472	10 259	1528	8415	4013.34
FD001→FD004	52 053	86 139	156 224	68 859	2698	5379	11 577	6860.69
FD002→FD001	3590	157 672	640	689	23 401	1150	351	345.03
FD002→FD003	23 071	19 053	62 823	11 029	12 756	3436	5213	1138.53
FD002→FD004	62 852	52 372	44 872	16 856	25 738	5150	15 106	5331.18
FD003→FD001	4581	18 307	25 826	32 451	2931	1487	1451	528.34
FD003→FD002	73 026	32 112	1 012 978	459 911	6754	4954	5257	14 831.46
FD003→FD004	11 407	296 061	275 665	82 520	5775	6888	3219	2944.50
FD004→FD001	154 842	45 394	162 100	43 794	13 377	4039	1840	2395.77
FD004→FD002	38 095	38 221	179 243	23 822	4927	8573	4460	3596.47
FD004→FD003	69 197	7977	1623	1117	1679	1440	682	597.53
Mean	40 851.33	72 700.33	184 667.75	64 865.42	9414.92	3896	4974.42	3877.68

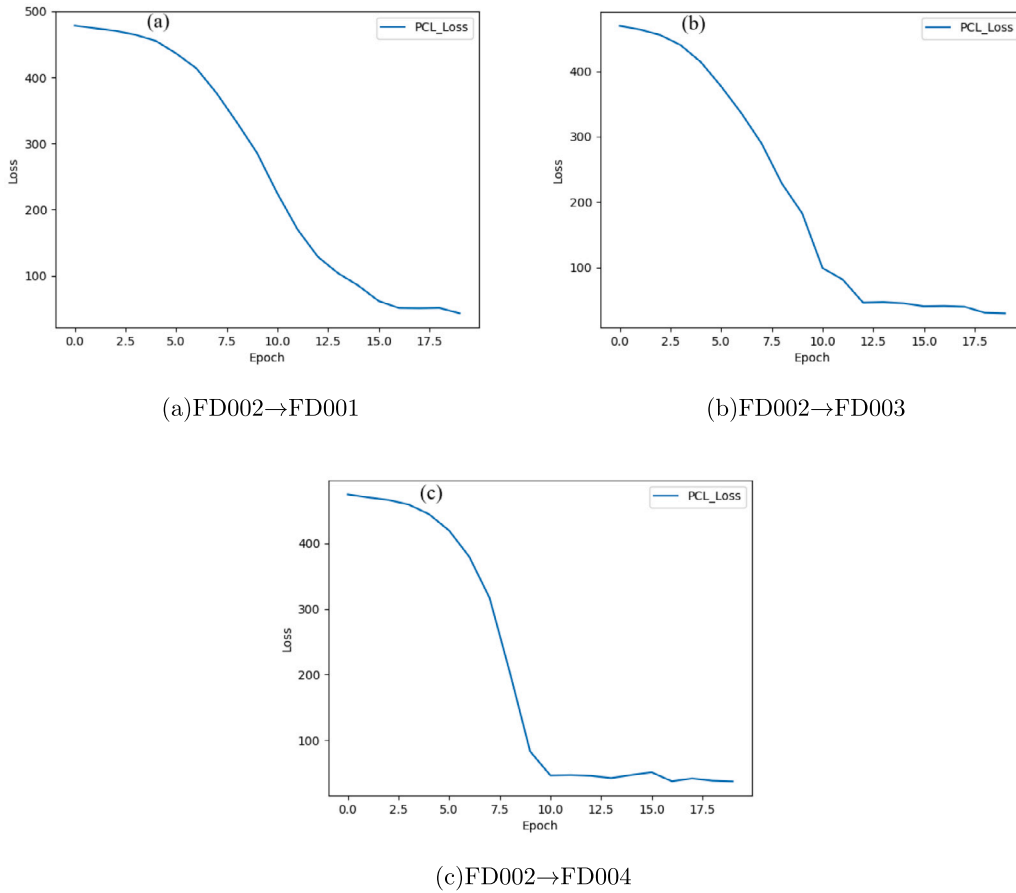


Fig. 11. PCL loss with source domain FD002.

performs better than that of FD001. Thus, the fault modes may contain more degradation information than operating conditions. For the transfer tasks when FD001 and FD003 are chosen as source domains, the number of fault modes may be a key factor affecting the performance of open set transfer. Researchers should not only consider operating conditions but also fault modes when studying cross domain RUL prediction especially in open set transfer.

3.5. The visual analysis of the network

In order to study the essential impact of LADA on features, this paper visualizes and analyzes the features and sensor data of the source and target domains. Specifically, taking FD001→FD002 as an example, Fig. 12 shows the distribution of mean, variance, and quartiles of the original data. Fig. 13 shows the distribution map of two domain features aligned by LADA.

From Fig. 12, there is a slight overlap in the data distribution of the two domains. Overall, there are significant differences in data levels. This also intuitively illustrates the challenge of cross domain RUL prediction. It is not feasible to directly apply the degradation information of the source domain to the target domain. From Fig. 13, it can be seen that the difference in feature distribution after LADA alignment is significantly reduced. The overlap between the two domains has been greatly improved. By comparing Figs. 12 and 13, it can be observed that the proposed method can effectively extract domain invariant features. This enables LADA to obtain satisfactory RUL prediction results.

4. Conclusion

In response to the difficulty in collecting degradation data of devices in practical scenarios and the different operating conditions, this paper proposes the LADA method by considering the transfer quality of source domain data and the full utilization of hidden label information in the target domain. During the prediction phase, a similarity measurement indicator and a pseudo class label construction method is introduced to enhance the transfer quality of source data and make the most use of hidden label information from target domain. Among them, Pearson correlation coefficient, dynamic time warping method, and desirability function are used to construct similarity measurement indicators. Cloud model theory and golden section method are used to construct pseudo class labels. The loss of feature differences is determined using the maximum mean difference and K-S statistical to minimize the disparity in feature distributions. Subsequently, an adversarial domain adaptation network for RUL prediction is developed and the network is trained together with pseudo class labels loss, feature difference loss and adversarial loss. This paper validates the effectiveness of the proposed method on the publicly available dataset C-MAPSS. Ablation experiments and comparative experiments were set up separately. The experimental results demonstrate the effectiveness of transfer learning, feature difference module, and PCL module. The comparative experimental results also demonstrate the satisfactory generalization ability of the LADA method.

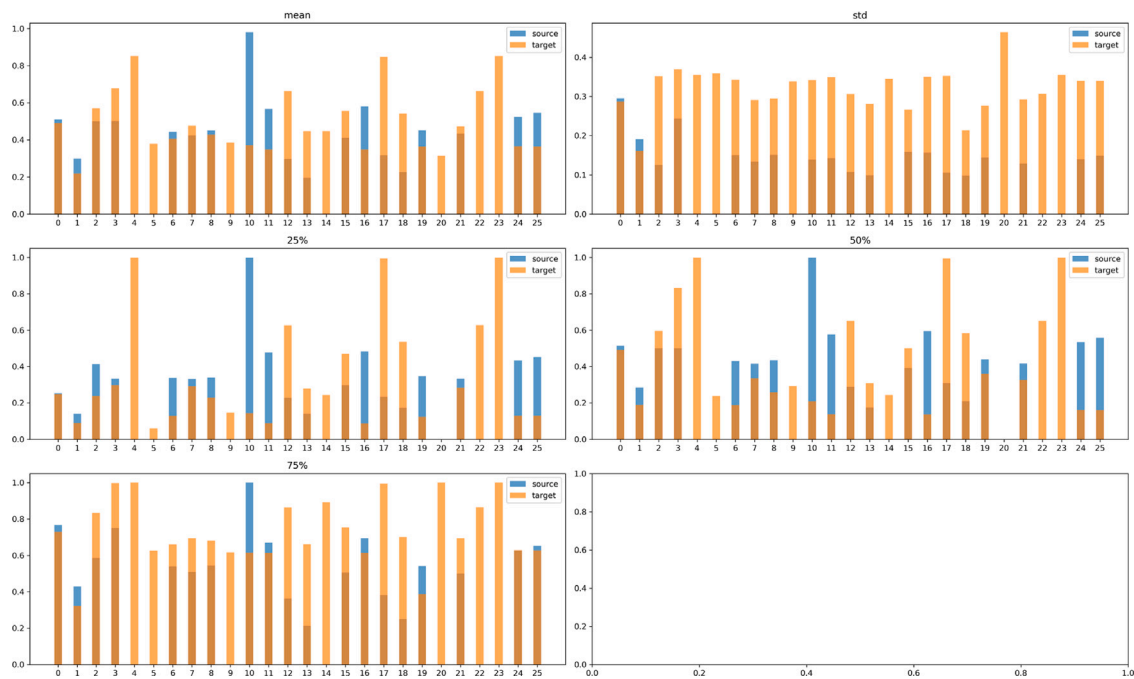


Fig. 12. Data distribution map of FD001→FD002.

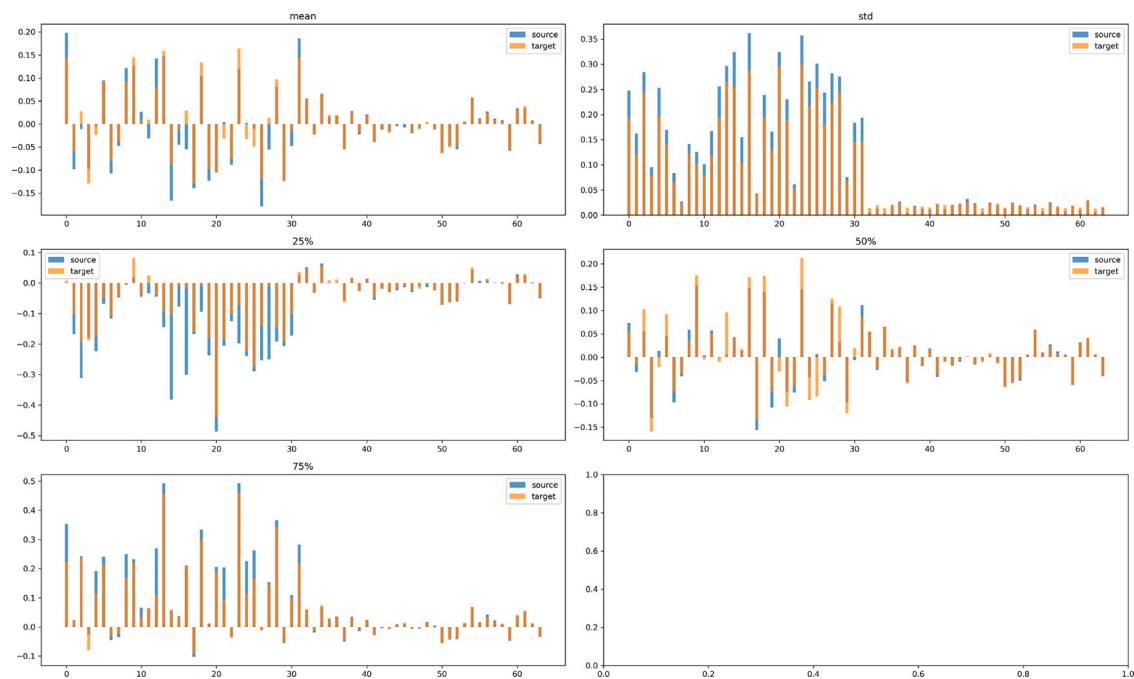


Fig. 13. Feature distribution map of FD001→FD002.

Due to the complexity of industrial systems, the operational conditions contained in the target domain data may include those that are not present in the source domain. Therefore, how to accurately predict RUL for target domain data in the absence of source domain data with similar distributions is a research worthy issue. In addition, how to predict the remaining useful life of the equipment while assessing its health status is also a problem that needs to be solved in the future.

CRediT authorship contribution statement

Shanshan Lv: Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Chengcheng Xia:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Cong Cheng:** Writing – review & editing, Formal analysis. **Jianhai Yan:** Writing – review & editing, Formal

analysis. **Xiaodan Wu:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Data availability

Data will be made available on request.

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